

Pharmaceutical Science

Science

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Short Title: Pharmaceutical Science
Faculty Science and Computing
Department Science
Cluster Pharmaceutical Science
Author JPOWER
Credits 5

Level: Introductory

Description of Module / Aims

This module gives an insight into the application of analytical science to pharmaceutical products, and the role of pharmacopoeias in the regulation of the safety and quality of pharmaceutical products. The module gives an overview of pharmaceutical production and the different types of drug formulations.

Programmes

SCIE-0002	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 3 / M
SCIE-0002	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 3 / M

Indicative Content

- Role of pharmacopoeias and ICH in pharmaceutical analysis and regulation; contents of typical monographs
- Overview of types of pharmaceutical formulations; detailed description of tablet production and typical tablet contents
- Structure, function and use of excipients in tablet formulations
- Introduction to statistical quality control in the pharmaceutical Industry; application to QC charts for defective products
- Operating curves and sampling in pharmaceutical analysis: single and double sampling plans
- Assay and identity testing of pharmaceuticals by pharmacopoeial methods; physical tests for tablets
- Structure and degradation of typical pharmaceutical products: reaction types and examples; stability studies and shelf-life prediction
- Information sources for pharmaceuticals; use of library and on line resources

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the role of pharmacopoeias in the pharmaceutical industry.
2. Use a Pharmacopoeial monograph to analyse drug formulations.
3. Describe the content and production of a typical tablet formulation.
4. Discuss, with examples, the stability of drug formulations.
5. Describe the application of statistical quality control in pharmaceutical production.
6. Use Pharmacopoeial procedures to identify and test pharmaceuticals.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Seminars.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,3,4,5
Continuous Assessment	50%	
<i>Practical</i>	50%	2,6

Assessment Criteria

<40%: No understanding of the basic concepts of pharmaceutical science.

40%-49%: Able to understand the basics of pharmaceutical science. May have some difficulty with clear descriptions.

50%-59%: Good level of achievement of learning outcomes.

60%-69%: In addition to the above, good critical thinking abilities. Shows a high level of competence and efficiency in relation to learning outcomes.

70%-100%: All previous to excellent level. Be able to describe and discuss aspects of pharmaceutical science and show initiative in approaching problems.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Khar, R. and S. Vyas. *Theory and Practice of Industrial Pharmacy*. 4th ed. London: CBS, 2013.
- Pharmaceutical, S. *British Pharmacopoeia 2015*. London: HMSO, 2015.

Requested Resources

- Room Type: Chemistry Lab